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Dipole-dipole interaction immobilizing residual solvent for stable polymer-based solid-state lithium batteries

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ABSTRACT

Poly(vinylidene fluoride) (PVDF)-based solid polymer electrolytes (SPEs) are regarded as promising candidates for solid-state lithium batteries due to their excellent flexibility and processability. However, their practical applications are hindered by insufficient ion transport and severe interfacial side reactions, which originate from the strong Li^+ coordination and intrinsic electrochemical instability of residual solvents such as *N*,*N*-dimethylformamide (DMF). To address these DMF-induced issues, we report a dipole-anchoring strategy that immobilizes residual DMF by integrating poly(1,3-dioxolane) (PDOL) into a poly(vinylidene fluoride-co-hexafluoropropylene) (PVHF)-based SPE matrix. The PDOL-DMF interaction weakens Li^+-DMF coordination and reshapes the solvation environment, resulting in an ionic conductivity of 0.62 mS cm^{-1} and an activation energy of 0.21 eV. Furthermore, this dipole-dipole interaction enhances the electrochemical stability of DMF toward electrodes through frontier molecular orbital modulation, thereby promoting the formation of inorganic-rich interphases on both lithium anodes and high-voltage cathodes. Consequently, Li||Li symmetric cells deliver ultra-stable cycling over 5000 h at 0.1 mA cm^{-2}, Li||$LiFePO_4$ full cells achieve 2500 cycles at 3 C, and Li||$LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ cells retain 82.5% capacity after 500 cycles at 1 C. This work demonstrates dipole-anchoring of residual solvent as an effective design principle for enhancing ion transport and interfacial stability in polymer-based solid-state lithium batteries.

1. Introduction

The global transition toward renewable energy and electrified transportation demands battery systems with both high energy density and inherent safety. Solid-state lithium batteries (SSLBs), which replace flammable liquid electrolytes with solid electrolytes (SEs), have emerged as a transformative technology that fundamentally addresses safety concerns while enabling the use of high-capacity lithium metal anodes to achieve energy densities beyond 500 Wh kg^{-1} [1–3]. The performance of SSLBs is critically influenced by the properties of SEs, including ionic conductivity and interfacial compatibility with electrodes. Among various SEs, solid polymer electrolytes (SPEs) are particularly attractive for scalable industrial application due to their flexibility, processability, cost-effectiveness, and favorable electrode interfacial contact [1,4–8]. Conventional SPEs, such as the most extensively studied poly(ethylene oxide) (PEO)-based SPEs, suffer from low room-temperature ionic conductivity ($<10^{-5}$ S cm^{-1}) and a narrow electrochemical stability window, which limit their practical application [9,10]. Recently, poly(vinylidene fluoride) (PVDF)-based SPEs, configured as a “salt-in-polymer” matrix with residual solvent, have gained increasing attention for achieving relatively high room-temperature ionic conductivity ($>10^{-4}$ S cm^{-1}) [11–13]. Unlike PEO, in PVDF-based SPEs, Li^+ coordinates with residual solvents such as *N*,*N*-dimethylformamide (DMF) to form $[Li(DMF)_x]^+$ ionic solvates, which

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migrate along the polymer chains for Li^+ conduction [12,14,15].

Despite its indispensable role in enabling ionic conduction, the residual DMF poses several critical challenges. First, DMF is thermodynamically unstable against lithium metal, inducing uncontrollable interfacial side reactions [16,17]. This leads to a heterogeneous, organic-rich solid electrolyte interphase (SEI) that fails to suppress dendrites and causes rapid Coulombic efficiency decay (Fig. 1a). Second, owing to its poor oxidation resistance, DMF decomposes readily at high-voltage cathodes, thereby exacerbating cathode electrolyte interphase (CEI) growth. Third, the strong Li^+-DMF coordination in PVDF-based SPEs forms tightly bound solvation shells that impose high migration barriers, which hinder further improvement in ionic conductivity and lead to sluggish interfacial reaction kinetics [18–20].

To address these challenges, various strategies have been explored, including inorganic filler modification [21–23], salt/solvent optimization [24,25], and molecular-architectural engineering [15,26]. Among them, the molecular-architectural approach is particularly promising because it can create efficient ion-transport channels and enhance interfacial stability by introducing functional polymer segments [27–29]. Crucially, the interaction between the introduced polymer

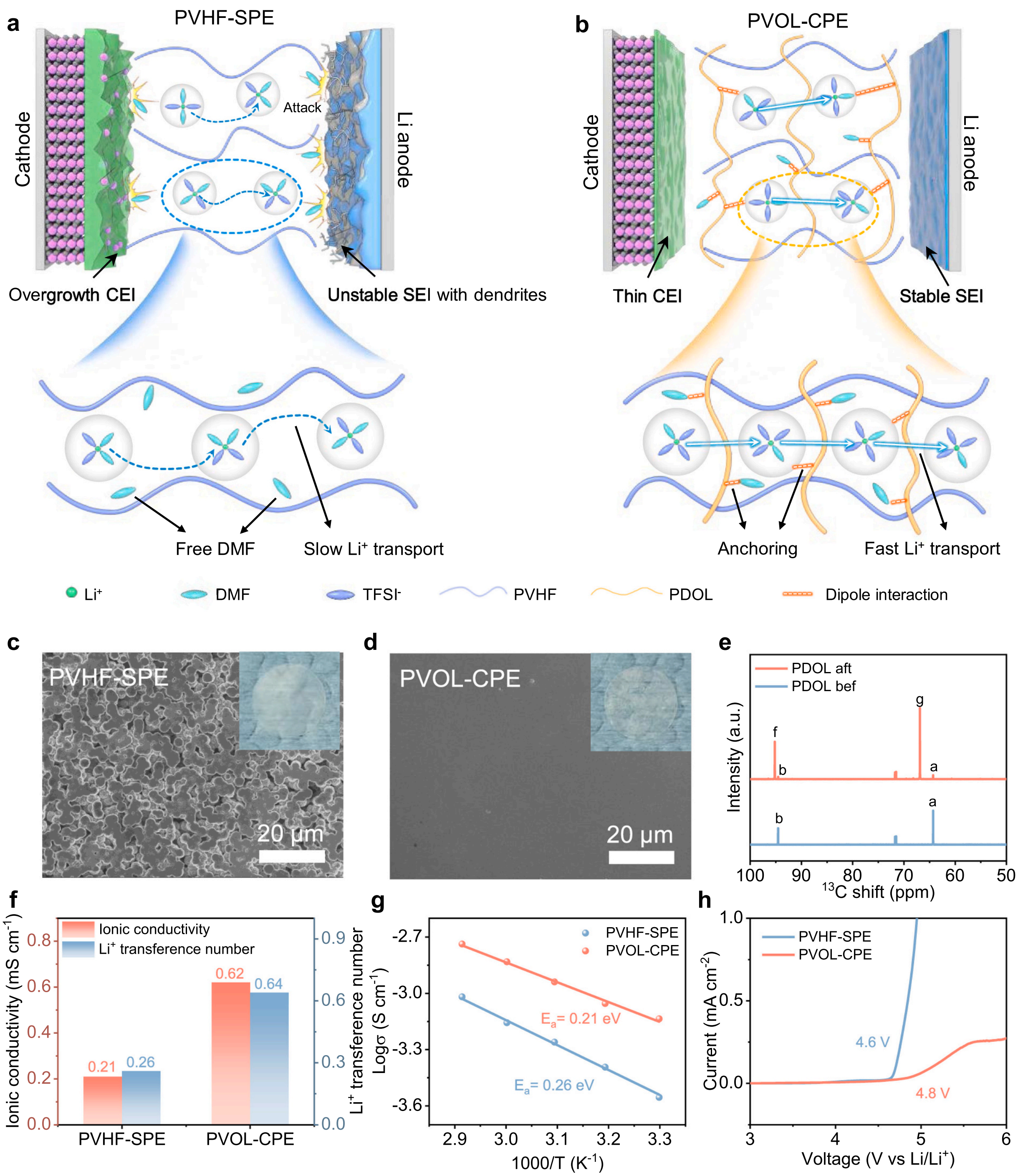

Fig. 1. (a) Schematic illustration of the dendrites formed on the Li-metal anode and degradation processes occurring on cathodes upon using PVHF-SPE. (b) Schematic illustration of the PVOL-CPE promoting ion transport and stabilizing the cathode and Li-metal anode interfaces. SEM images of (c) PVHF-SPE and (d) PVOL-CPE (inset: optical images of two electrolyte membranes). (e) ^{13}C NMR spectroscopy of PDOL before and after polymerization. (f) Ionic conductivity and Li^+ transference number of PVHF-SPE and PVOL-CPE. (g) Temperature-dependent ionic conductivities, and (h) LSV curves of the two electrolytes.

component and the residual solvent directly shapes the Li^+ solvation structure. However, deliberately exploiting such intermolecular interactions, for instance, dipole-dipole interactions, to immobilize the residual solvent has been rarely investigated.

Herein, we report a dipole-anchoring strategy that immobilizes residual DMF by integrating poly(1,3-dioxolane) (PDOL) into a poly (vinylidene fluoride-co-hexafluoropropylene) (PVHF)-based SPE matrix via in situ polymerization, forming a composite polymer electrolyte (PVOL-CPE). In the resulting electrolyte, the strong dipole interactions between PDOL and DMF play dual roles. First, they weaken Li^+-DMF coordination, which enhances the ionic conductivity to 0.62 mS cm^{-1} and reduces the activation energy to 0.21 eV, enabling rapid Li^+ transport. Second, they modulate the frontier molecular orbitals of DMF, which suppresses undesired interfacial side reactions and induces the formation of inorganic-rich interphases—a robust, Li^+-conductive SEI on the anode and a LiF-rich CEI on the high-voltage cathode (Fig. 1b). As a result, Li||Li symmetric cells achieve ultra-stable cycling for over 5000 h at 0.1 mA cm^{-2} and 0.1 mAh cm^{-2}, and full cells with $LiFePO_4$ (LFP) cathodes exhibit remarkable cycling stability over 2500 cycles at 3 C at room temperature. Furthermore, the developed electrolyte retains 82.5% capacity after 500 cycles at 1 C when paired with high-voltage $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM811) cathodes. This work demonstrates that dipole anchoring transforms a detrimental residual solvent into a stabilizing component, providing a general and powerful strategy for achieving high-energy-density solid-state lithium batteries.

2. Results and discussion

A precursor solution of 1,3-dioxolane (DOL) was infiltrated into a fluoropolymer backbone (PVHF-SPE). Subsequent in situ cationic ring-opening polymerization of DOL produced PDOL as the polyether component within the fluoropolymer matrix, forming the composite polymer electrolyte PVOL-CPE (Figure S1). The ring-opening polymerization mechanism is that LiDFOB first decomposes to BF_3, which then reacts with residual water to generate a protonic acid, and this protonic acid protonates the oxygen atom of DOL, triggering ring opening and subsequent polymerization. Scanning electron microscopy (SEM) revealed that the pristine PVHF-SPE possesses a porous morphology composed of interconnected spherulites (Fig. 1c). After PDOL incorporation, the PVOL-CPE shows a smooth surface morphology (Fig. 1d), indicating good compatibility between the components. The thickness of the PVOL-CPE membrane, as determined by the cross-sectional SEM, is ~55 μm, comparable to that of the PVHF-SPE membrane (Figure S2). The thin polymer electrolyte is necessary for achieving high-energy-density SSLBs [30]. The polymerization efficacy of the PDOL was confirmed by the transition from a liquid precursor to a solid, non-fluid material (Figure S3). The successful polymerization of the PDOL was further verified by Fourier-transform infrared (FTIR) spectroscopy and nuclear magnetic resonance (NMR) spectroscopy. The characteristic peak of DOL monomer at 912 cm^{-1} (C-H out-of-plane bending vibration) disappears and is replaced by a new peak at 848 cm^{-1} corresponding to long-chain PDOL (Figure S4). ^{13}C NMR spectroscopy showed a pronounced decrease in DOL monomer peaks (a, b) and the emergence of new signals (f, g) assigned to the PDOL polymer backbone (C-C-O and O-C-O bonds), confirming chain formation of PDOL (Fig. 1e). This was also supported by ^{1}H NMR spectroscopy (Figure S5). Thermogravimetric analysis (TGA) demonstrated that PVOL-CPE undergoes initial weight loss before 200°C due to moisture, residual DMF, and PDOL degradation [31], followed by lithium salt decomposition up to 320°C (Figure S6) [24]. The primary polymer decomposition occurs near 420°C for PVHF powder, but this temperature is lowered for both PVHF-SPE and PVOL-CPE due to the interaction between the polymer matrix and lithium salt [14]. Furthermore, the XRD patterns of PVHF powder show peaks at 18.0° and 26.7° assigned to the (100) and (021) planes of the α phase, and a peak at 20.2° assigned to the (110)/(200) planes of the β phase (Figure S7). In the PVHF-SPE and PVOL-CPE membranes, the α-phase peaks are significantly weakened, indicating markedly reduced PVDF crystallinity, which facilitates Li^+ conduction.

The ionic conductivity and Li^+ transference number (t_{Li^+}) are critical metrics for evaluating Li^+ transport in polymer electrolytes. By measuring the ionic conductivity of composite electrolytes with different added amounts of PDOL precursor solution, the optimal amount was determined to be 20 μL (Figure S8). The corresponding PVOL-CPE exhibits an ionic conductivity of 0.62 mS cm^{-1}, surpassing that of the PVHF-SPE (0.21 mS cm^{-1}) (Fig. 1f). Furthermore, PVOL-CPE achieves a high t_{Li^+} of 0.64, significantly greater than the value for PVHF-SPE (0.26) (Figure S9), which is beneficial to mitigating concentration polarization [32]. The enhanced ion transport kinetics are corroborated by a lower activation energy for Li^+ migration in PVOL-CPE (0.21 eV) compared to PVHF-SPE (0.26 eV), as derived from temperature-dependent impedance measurements (Fig. 1g). To assess electrochemical stability, linear sweep voltammetry (LSV) was performed. PVOL-CPE demonstrates an expanded electrochemical stability window of 4.8 V, higher than the 4.6 V observed for PVHF-SPE (Fig. 1h), indicating its potential compatibility with high-voltage cathodes.

FTIR and Raman spectroscopy analyses confirm a dipole-dipole interaction between the PDOL and residual DMF solvent molecules in the PVOL-CPE. In the FTIR spectra (Fig. 2a), the characteristic peaks at 1654 cm^{-1} (C=O) and 1390 cm^{-1} ($-CH_3$) of residual DMF in the PVHF-SPE shift to 1658 cm^{-1} and 1402 cm^{-1}, respectively, in the PVOL-CPE. Concurrently, the C-O-C stretching vibration of PDOL at 1830 cm^{-1} undergoes a blue shift to 1835 cm^{-1}. These collective shifts provide direct evidence of a strong intermolecular interaction between PDOL and residual DMF. Density functional theory (DFT) calculations substantiate this interaction. Electrostatic potential (ESP) analysis reveals that the electronegative oxygen atom of PDOL's ether group possesses a localized negative potential, while the methyl terminus of DMF exhibits a significant positive potential (Fig. 2b), creating a favorable alignment for a strong dipole-dipole attraction. Furthermore, the calculated binding energy for the PDOL-DMF complex (-0.73 eV) is significantly greater than that of the DMF-DMF dimer (-0.09 eV), confirming that PDOL can effectively anchor DMF molecules by the dipole-dipole interaction (Fig. 2c). This dipole anchoring fundamentally alters the chemical reactivity at the electrode interface. Frontier molecular orbital analysis shows the lowest unoccupied molecular orbital (LUMO) energy of DMF (0.60 eV) is lower than that of PDOL (0.85 eV), indicating DMF's higher susceptibility to reduction by lithium metal (Fig. 2d). Upon complexation with PDOL, the LUMO energy of the anchored DMF increases, directly reflecting its enhanced electrochemical stability against lithium metal anode. In addition, the highest occupied molecular orbital (HOMO) energy of the anchored DMF decreases, which also explains the improved oxidative stability.

Critically, this PDOL-DMF interaction can modulate the Li^+ solvation structure. The Raman peak near 671 cm^{-1} in the PVHF-SPE, assigned to Li^+ coordinated to the carbonyl oxygen (O=C-N) of DMF, redshifts to 668 cm^{-1} in the PVOL-CPE (Fig. 2e). This shift signifies a weakened ion-dipole interaction between DMF and Li^+, a finding corroborated by FTIR analysis (Figure S10). The dipole-anchoring-induced weakening of Li^+-DMF coordination effectively lowers the desolvation energy barrier, thereby facilitating rapid Li^+ transport. Concurrently, this disruption of the primary solvation structure strengthens the interaction between Li^+ and anions, as confirmed by Raman spectroscopy (Fig. 2f). While both PVHF-SPE and PVOL-CPE primarily consist of contact ion pairs (CIP) and aggregates (AGG), with no detectable solvent-separated ion pairs (SSIP), quantitative analysis reveals a significantly higher proportion of AGG in PVOL-CPE compared to PVHF-SPE (Fig. 2g). This anion-rich AGG solvation structure facilitates Li^+ hopping via a low-barrier pathway while restricting anion migration, thereby accounting for the improved Li^+ transference number [33–35]. Moreover, it suppresses excessive DMF decomposition and promotes the formation of an inorganic-rich SEI. This altered coordination environment was also verified by solid-state nuclear magnetic resonance (SSNMR). The ^{7}Li

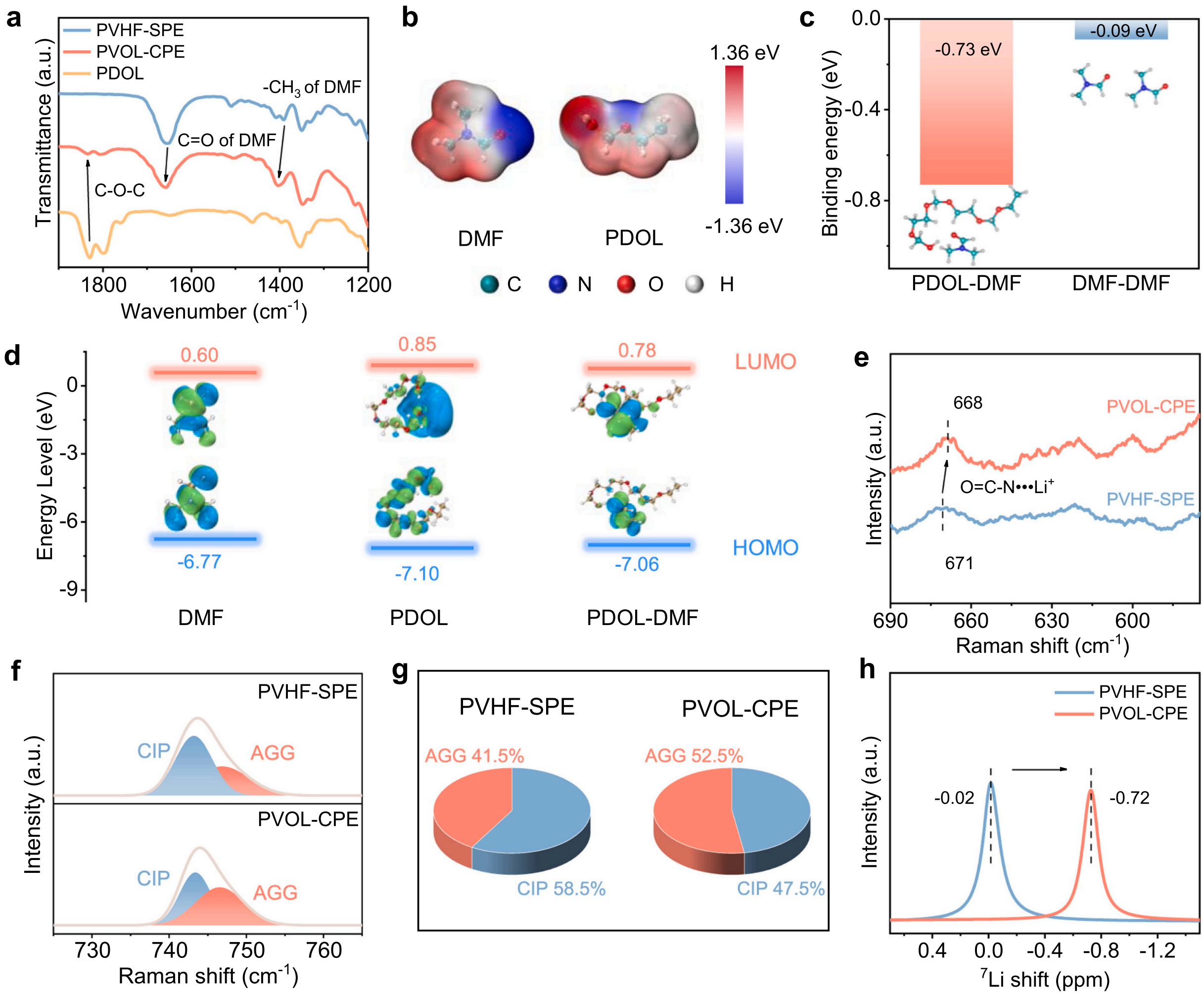

Fig. 2. (a) FTIR spectroscopy of PDOL, PVHF-SPE, and PVOL-CPE. (b) Electrostatic potential distributions of DMF and PDOL molecules. (c) Binding energy of the PDOL-DMF complex and DMF-DMF dimer. (d) LUMO and HOMO levels of the PVOL-CPE components. Raman spectroscopy of (e) Li^+-DMF interactions and (f) Li^+-anion coordination in PVHF-SPE and PVOL-CPE. (g) Comparison of the content of different solvation structure configurations for the two electrolytes. (h) ^7Li SSNMR of PVHF-SPE and PVOL-CPE.

signal in the PVOL-CPE shifts upfield relative to that in PVHF-SPE, indicating an enhanced shielding effect on Li^+. This change is attributed to the intensified coordination between Li^+ and anions, a conclusion that aligns perfectly with the Raman spectroscopy data.

The cycling reversibility of the lithium metal anode was evaluated by measuring the Coulombic efficiency (CE) of Li||Cu half-cells. Cells employing the PVOL-CPE sustained a high average CE of ~90% over 500 cycles, significantly outperforming those with the PVHF-SPE (Figure S11). This enhanced reversibility is supported by faster interfacial kinetics: the exchange current density (j_0) at the Li/PVOL-CPE interface is 0.69 mA cm^{-2}, approximately 27 times greater than that of the PVHF-SPE interface (0.025 mA cm^{-2}) (Fig. 3a). Furthermore, the activation energy for Li^+ transport through the SEI film (E_{SEI}), derived from Arrhenius analysis of temperature-dependent electrochemical impedance spectra (EIS), is lower for PVOL-CPE (35.69 kJ mol^{-1}) than that for PVHF-SPE (48.79 kJ mol^{-1}), indicating more facile Li^+ diffusion across the interface (Fig. 3b). Finally, critical current density (CCD) tests demonstrate the superior dendrite suppression capability of PVOL-CPE, which withstands a current density of 2.7 mA cm^{-2} before short-circuiting, vastly exceeding the 0.7 mA cm^{-2} CCD of PVHF-SPE (Fig. 3c).

The reversibility of Li^+ plating/stripping was evaluated in Li||Li symmetric cells. The Li||Li symmetric cells with the PVHF-SPE exhibit large overpotentials and a short cycling life of 60 h at 0.1 mA cm^{-2} and 0.1 mAh cm^{-2}, attributable to uncontrolled side reactions and Li dendrite growth (Fig. 3d). In sharp contrast, cells with the PVOL-CPE demonstrated stable cycling for 5000 h with a low overpotential of ~35 mV under the same conditions. At a higher current density of 0.3 mA cm^{-2} and a capacity of 0.3 mAh cm^{-2}, the Li||PVHF-SPE||Li cell displays a failure after only 20 h, while the Li||PVOL-CPE||Li cell shows a much longer cycling life of 700 h (Figure S12). This performance advantage persisted under more demanding conditions: at 1 mA cm^{-2} and 1 mAh cm^{-2}, PVOL-based cells cycle stably for over 450 h (Fig. 3e), indicating the excellent ability of PVOL electrolyte for Li dendrite suppression. In addition, EIS measurements of Li||Li symmetric cells show a lower interfacial impedance of cells with PVOL-CPE before cycling, suggesting the improved electrode/electrolyte contact achieved by in situ polymerization (Figure S13). Ex-situ SEM of the cycled Li anode revealed stark morphological differences. The PVHF-SPE leads to protrusion-like, high-surface-area Li deposits susceptible to severe corrosion. In contrast, PVOL-CPE enables dense and flat Li deposition (Fig. 3f). Cross-sectional analysis confirmed a compact and thin Li deposition layer (2 μm) with PVOL-CPE, in contrast to a thick, porous layer (8 μm) formed with PVHF-SPE (Fig. 3g). Atomic force microscopy (AFM) further quantified this contrast: Li deposits with PVHF-SPE exhibit a rough texture with an average roughness (R_a) of 293 nm, while those with PVOL-CPE are remarkably flat (R_a = 18 nm) (Fig. 3h). Furthermore, we employed distribution of relaxation times (DRT) analysis to probe interfacial evolution and kinetics (Fig. 3i; Figure S14).

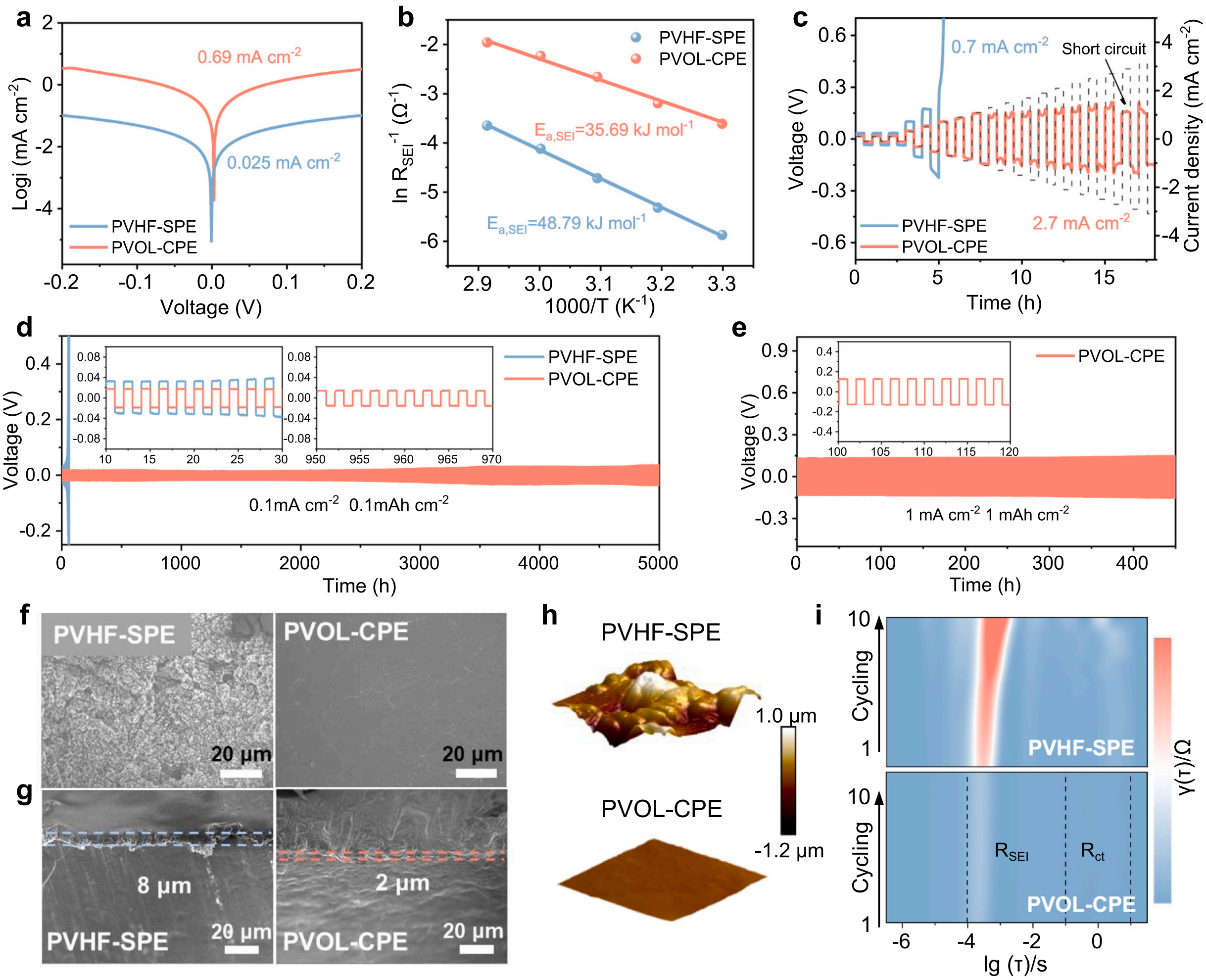

Fig. 3. (a) Tafel plots for Li deposition/stripping of the Li||Li symmetric cells with different electrolytes. (b) Activation energy of the transport of Li^+ across SEI with different electrolytes. (c) CCD test of Li||Li symmetric cells with different electrolytes. (d) Galvanostatic cycling curves of Li||Li cells using PVHF-SPE and PVOL-CPE at a current density of 0.1 mA cm^{-2} with 0.1 mAh cm^{-2} capacity. (e) Galvanostatic cycling curves of Li||Li cells using PVOL-CPE at a current density of 1 mA cm^{-2} with 1 mAh cm^{-2} capacity. (f) Top-view and (g) cross-sectional view of SEM images for cycled Li anodes at 0.1 mA cm^{-2} and 0.1 mAh cm^{-2} with different electrolytes. (h) AFM images of cycled Li anode with different electrolytes. (i) DRT results of Li||Li symmetric cells during cycling with different electrolytes.

Two primary electrochemical processes were identified at 10^{-4}-10^{-1} s and 10^{-1}-10 s, corresponding to SEI resistance (R_{SEI}) and charge-transfer resistance (R_{ct}), respectively [36,37]. For the PVHF-based cell, the R_{SEI} value increases continuously during cycling, indicating persistent interfacial parasitic reactions. In contrast, the PVOL-based cell exhibits exceptional cycling stability with minimal fluctuation in R_{SEI}. In addition, the R_{ct} value is consistently lower with PVOL-CPE, signifying faster interfacial charge-transfer kinetics. The cycled electrolyte membranes were further examined. PVOL-CPE remains smooth and transparent, confirming its excellent stability during cycling (Figure S15a). In contrast, PVHF-SPE turns dark brown due to PVDF dehydrofluorination, and its SEM image reveals numerous side reaction products deposited on the membrane (Figure S15b).

The chemical composition and spatial distribution of the SEI film were investigated using X-ray photoelectron spectroscopy (XPS) and time-of-flight secondary ion mass spectrometry (TOF-SIMS). The F 1 s spectra (Fig. 4a, b) show two primary peaks at 685.0 and 688.7 eV, corresponding to LiF and C-F species, respectively [38]. For both electrolytes, the SEI surface is dominated by organic C-F species. However, depth profiling reveals a critical divergence that underscores the effect of dipole anchoring: the LiF content increases with sputtering depth in the PVOL-derived SEI, whereas it decreases in the PVHF-derived SEI. Consequently, the ratio of LiF to organic fluorine species remains consistently higher throughout the entire depth in PVOL-CPE system, confirming a LiF-dominated SEI composition. This LiF enrichment supports a shift in interphase chemistry induced by PDOL-DMF anchoring. By weakening Li^+-DMF coordination and enhancing Li^+-anion association, the localized solvation environment favors inorganic interphase formation and suppresses solvent-derived organic decomposition. The distribution of nitrogen species also differs markedly (Figure S16). The PVHF-derived SEI is primarily composed of N-S species, while the PVOL-derived SEI shows a higher content of inorganic nitrogen species, including a distinct Li_3N peak. More importantly, the C 1 s spectra indicate lower concentrations of C-N, C-O, and C=O species in the PVOL-derived SEI (Fig. 4c,d; Figure S17), directly evidencing suppressed decomposition of residual DMF as a result of the dipole-anchoring effect of PDOL [39]. The results from TOF-SIMS align well with the XPS findings. As shown in Fig. 4e and Figure S18, the TOF-SIMS 3D spectra show an abundant and uniform distribution of LiF fragments in the PVOL-derived SEI. However, the SEI with PVHF-SPE displays an insufficient presence of LiF and exhibits a non-uniform distribution. The PVOL-derived SEI also exhibits higher signals for Li_3N and lower signals for Li_2CO_3, indicating the formation of an interface conducive to fast Li^+ transport and improved stability, as Li_3N is a known Li^+ conductor while Li_2CO_3 can perpetuate parasitic reactions [40–42]. Critically, a thin surface layer of C_2H_6O, a DMF decomposition product [12], is detected in the PVOL-CPE system, whereas a thick, organic-rich layer permeates the PVHF-derived SEI. Collectively, these results confirm that the

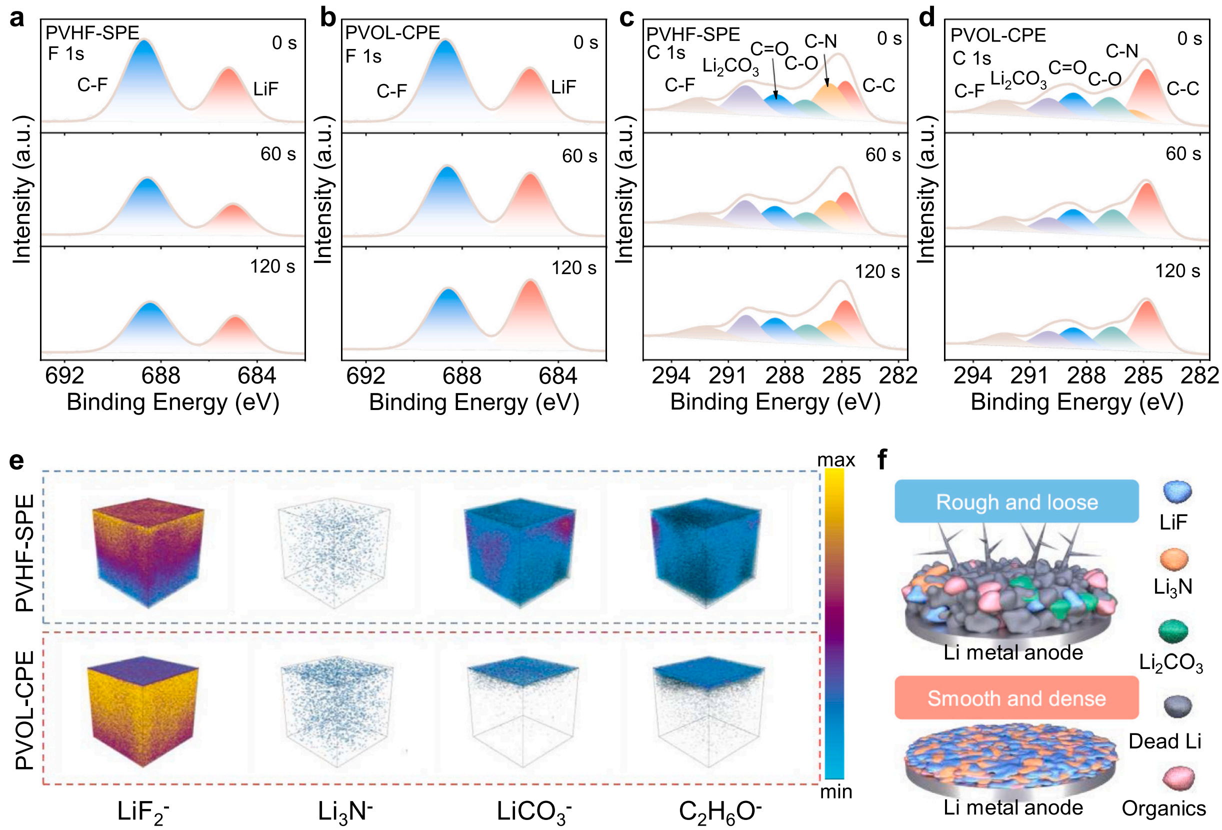

Fig. 4. The XPS depth profiles of F 1 s with (a) PVHF-SPE and (b) PVOL-CPE for cycled Li anode. The XPS depth profiles of C 1 s with (c) PVHF-SPE and (d) PVOL-CPE for cycled Li anode. (e) TOF-SIMS 3D spectra of LiF_2^-, Li_3N^-, $LiCO_3^-$, and $C_2H_6O^-$ fragments for cycled Li anode with different electrolytes. (f) Schematic diagram of SEI derived from different electrolytes.

dipole-anchoring strategy effectively immobilizes DMF, suppresses its decomposition, and induces an anion-driven formation of a stable, inorganic-rich SEI dominated by LiF and Li_3N (Fig. 4f).

To assess the practical viability of the PVOL-CPE, we assembled full cells with a lithium metal anode paired with $LiFePO_4$ (LFP) or $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM811) cathodes. The Li||PVOL-CPE||LFP cells show an exceptional cycling stability with a capacity retention of 95.6% after 300 cycles at 0.5 C, whereas cells with PVHF-SPE begin rapid decay after only 60 cycles (Figure S19). The EIS spectra (Figure S20) of Li||LFP full cells before and after cycling indicate that during cycling, the impedance of the cell with PVHF-SPE gradually increases, whereas that of the cell with PVOL-CPE gradually stabilizes, suggesting that PVOL-CPE induces the formation of stable interfaces at both the cathode and anode. Long-term cycling at a high rate of 3 C further shows Li||PVOL-CPE||LFP cells retain 72.1% capacity after 2500 cycles and room temperature with an average Coulombic efficiency of 99.9% (Fig. 5a). This contrasts sharply with Li||PVHF-SPE||LFP cells, which show an extremely low discharging specific capacity of 10.3 mAh g^{-1} after 140 cycles. In rate capability tests (Fig. 5b), Li||PVOL-CPE||LFP cells deliver discharge specific capacities of 167.6, 167.4, 159.9, 147.2, 130.1, and 113.7 mAh g^{-1} at 0.1, 0.2, 0.5, 1, 2, and 3 C, respectively, significantly outperforming PVHF-based cells. The corresponding charge and discharge curves at different current densities reveal lower voltage polarization with PVOL-CPE (Fig. 5c; Figure S21). Notably, when paired with high-mass-loading LFP cathodes (12.3 mg cm^{-2}), Li||PVOL-CPE||LFP cells maintain a high discharging specific capacity of 154.8 mAh g^{-1} at 0.5 C for over 200 cycles (Fig. 5d). The applicability of PVOL-CPE was further verified in high-voltage systems: Li||PVOL-CPE||NCM811 cells exhibit an excellent capacity retention of 82.5% after 500 cycles at 1 C and room temperature, while PVHF-based cells fail within 150 cycles (8.9%) (Fig. 5e). PVOL-based cells also show superior rate performance with NCM811 cathodes (Figure S22). The EIS spectra of Li||NCM811 full cells before and after cycling at 1 C (Figure S23) also confirm the stability of PVOL-CPE toward the high-voltage cathode. This outstanding performance across multiple cell configurations is directly attributed to the dipole-anchoring effect of PDOL on residual DMF, which simultaneously facilitates rapid Li^+ transport and stabilizes both electrode interfaces. Therefore, our developed batteries exhibit superior lifespan compared with recently reported PVDF-based SSLBs, [13,15,21,25,32,43–46] especially at a high rate (Fig. 5f, Table S1).

To elucidate the mechanism underlying the exceptional cycling stability of high-voltage NCM811 cathodes with the PVOL-CPE, we performed a suite of electrochemical and interfacial characterization techniques. Repeated cyclic voltammetry (CV) curves at 0.05 mV s^{-1} for the Li||NCM811 cell with PVOL-CPE exhibit three pairs of well-defined redox peaks after 10 cycles, revealing highly reversible phase transition behavior (Fig. 6a; Figure S24). The first pair corresponds to the hexagonal H1 to monoclinic M phase transition accompanied by Ni^{2+}/Ni^{3+} redox, the second pair to the M to hexagonal H2 transition involving Ni^{3+}/Ni^{4+} redox, and the third pair to the H2 to hexagonal H3 transition. The excellent retention of these peaks indicates that dipole anchoring effectively inhibits the oxidative decomposition of DMF at the cathode interface. Galvanostatic intermittent titration technique (GITT) profiles after 30 cycles further confirm enhanced reaction kinetics, as evidenced by the consistently lower overpotential in cells with PVOL-CPE (Fig. 6b; Figure S25). This improvement is attributed to the weakened Li^+-DMF coordination caused by dipole anchoring, which facilitates interfacial charge transfer and reduces desolvation barriers. The magnitude of parasitic reactions at the cathode-electrolyte interface was quantified by holding cells at elevated voltages. Leakage current tests at 4.3, 4.4, and 4.5 V show that PVHF-based cells exhibit a substantial increase in current with rising voltage, whereas PVOL-based cells

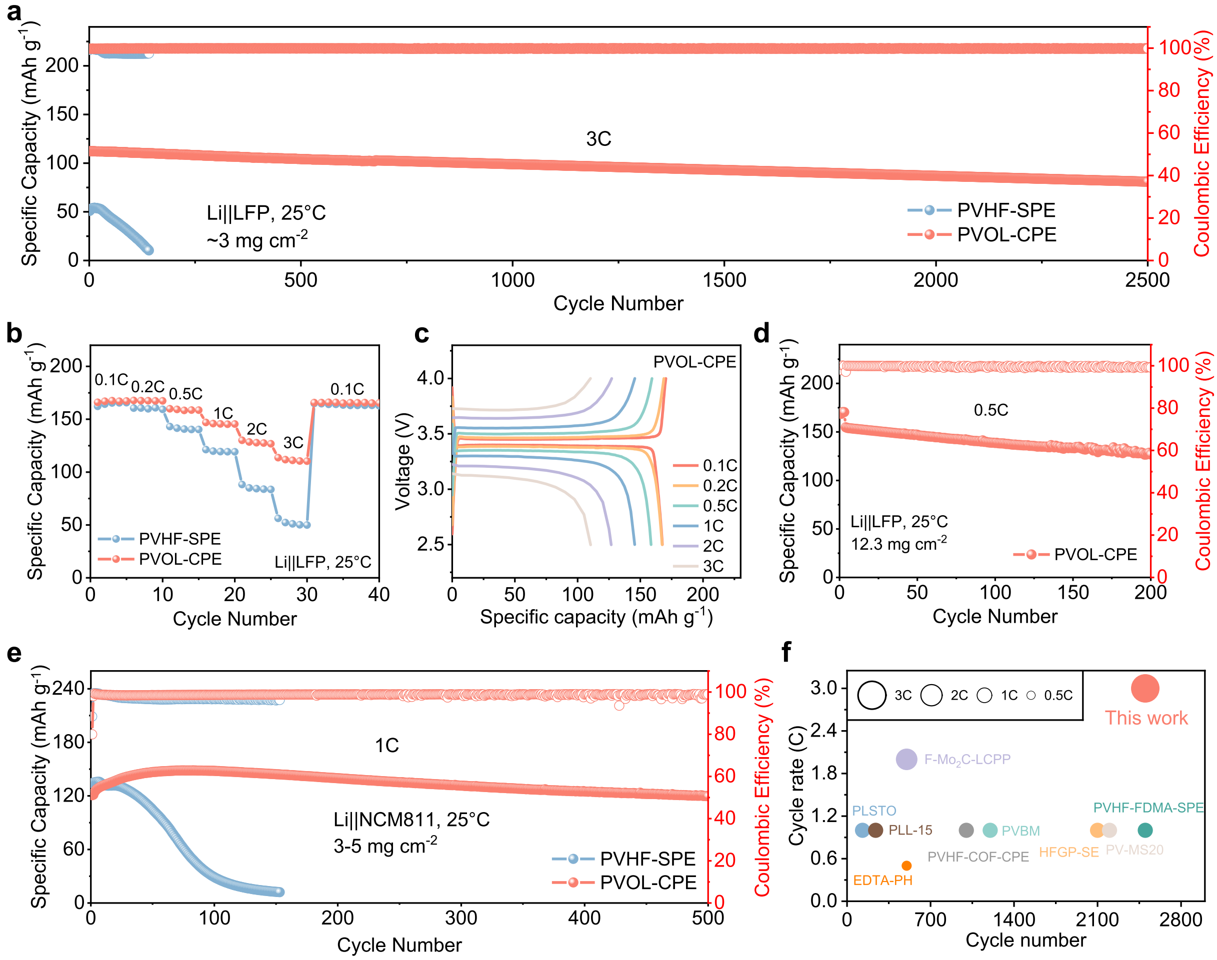

Fig. 5. (a) Long-term cycling performance of Li||LFP cells at 3 C. (b) Rate performance of Li||LFP cells at the current densities from 0.1 C to 3 C with different electrolytes. (c) The charge and discharge curves of Li||LFP cells at various current densities with PVOL-CPE. (d) Cycling performance of Li||LFP cells with high-mass-loading cathode (12.3 mg cm^{-2}) at 0.5 C. (e) Cycling performance of Li||NCM811 cells at 1 C with different electrolytes. (f) Electrochemical performance comparison of Li||LFP cells between this work and previously reported work.

maintain smaller, stable currents (Fig. 6c). This stark contrast demonstrates that dipole anchoring intrinsically enhances the oxidative stability of DMF by lowering its HOMO energy level, thereby inhibiting high-voltage decomposition. Furthermore, in situ EIS was employed to track interfacial impedance evolution during delithiation and lithiation (Fig. 6d). At every equivalent voltage point, cells with PVOL-CPE exhibit lower impedance than those with PVHF-SPE. The impedance remains more stable and symmetric throughout the cycling process for PVOL-CPE due to the formation of a durable CEI mediated by the dipole-anchoring effect.

Post-cycling SEM revealed stark morphological differences in the NCM811 cathodes. Particles cycled in the PVHF-SPE show severe surface degradation, being encapsulated by entangled decomposition products and exhibiting pronounced fractures indicative of structural collapse (Fig. 6e). In contrast, cathodes from the PVOL-CPE maintain clean surfaces and intact particle morphology (Fig. 6f), demonstrating preserved structural integrity and effective suppression of stress-corrosion cracking [47]. Transmission electron microscopy (TEM) further elucidated the discrepancy of CEI formed within different electrolytes. After 50 cycles, the CEI in the PVHF-based cell is overgrown with a thickness of ~18 nm (Fig. 6g; Figure S26a), whereas it is thin and uniform with PVOL-CPE (~2 nm) (Fig. 6h; Figure S26b). This compact CEI directly reflects the ability of dipole anchoring to prevent runaway DMF decomposition and simultaneously foster an anion-derived inorganic interphase that facilitates fast Li^+ transport. XPS analysis confirmed the distinct chemical signatures. The CEI derived from PVOL-CPE exhibits lower intensities of C-O and C═O species, known oxidative decomposition products of DMF [12], compared to that from PVHF-SPE (Fig. 6i; Figure S27), confirming the suppressed DMF decomposition induced by dipole anchoring. Furthermore, the PVHF-derived CEI is dominated by organic fluorine species (C-F), while the PVOL-derived CEI is characterized by higher LiF content, confirming an inorganic-rich structure (Fig. 6j). These results demonstrate that the dipole-anchoring strategy endows the electrolyte with superior oxidation stability and the ability to form a protective, LiF-rich CEI. By immobilizing DMF and reshaping the interfacial chemistry, dipole anchoring effectively passivates the high-voltage cathode surface and inhibits continuous side reactions, thereby ensuring outstanding cycling performance.

3. Conclusion

In summary, we have demonstrated a dipole-anchoring strategy that addresses the dual challenges of limited Li^+ transport and interfacial instability in PVDF-based SPEs. By integrating PDOL into a PVHF matrix via in situ polymerization, we constructed a composite polymer electrolyte (PVOL-CPE) in which strong dipole-dipole interactions serve as molecular anchors to immobilize DMF. This dipole anchoring weakens Li^+-DMF coordination and reshapes the solvation environment, thereby facilitating rapid Li^+ transport. Furthermore, through modulating the

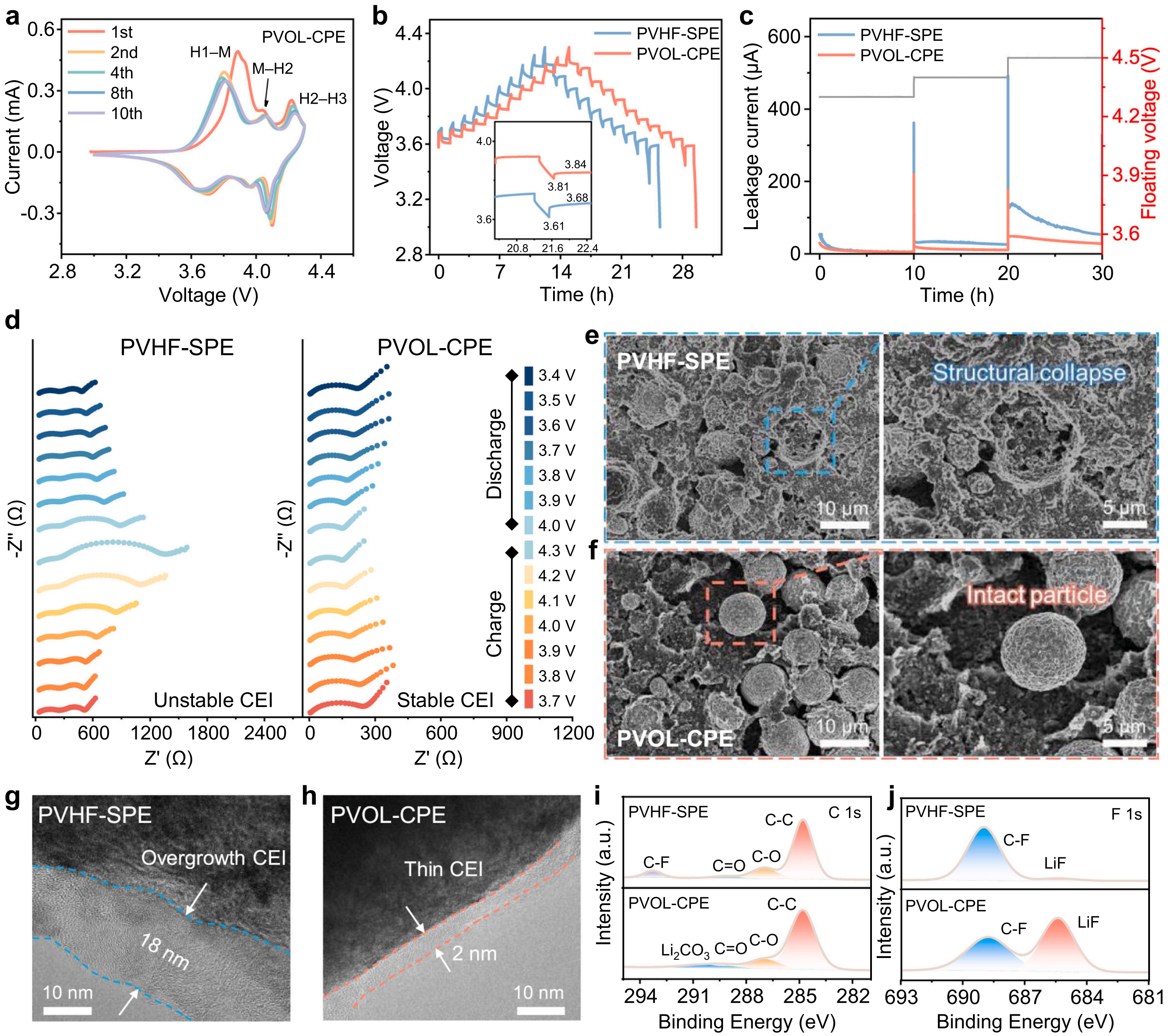

Fig. 6. (a) CV curves at 0.05 mV s^{-1} of Li||NCM811 full cells with PVOL-CPE. (b) Charge and discharge voltage curves of Li||NCM811 full cells after 30 cycles measured by GITT with the two electrolytes. (c) Floating voltage test of different electrolytes under 4.3, 4.4, and 4.5 V. (d) In situ EIS plots of Li||NCM811 full cells using different electrolytes after 80 cycles. SEM images of cycled NCM811 cathodes within (e) PVHF-SPE and (f) PVOL-CPE. HRTEM images of NCM811 cathodes cycled in (g) PVHF-SPE and (h) PVOL-CPE. (i) C 1 s and (j) F 1 s XPS spectra of cycled NCM811 cathodes using different electrolytes after 50 cycles.

LUMO and HOMO levels of residual DMF, this dipole interaction enhances its electrochemical stability, effectively inhibiting interfacial side reactions and promoting the formation of robust, inorganic-rich interphases on both the lithium anode and the high-voltage cathode. As a result, Li||Li symmetric cells with PVOL-CPE achieve unprecedented cycling stability exceeding 5000 h (0.1 mA cm^{-2}) and endure over 450 h even at a higher current density of 1 mA cm^{-2}. Full cells with LFP cathodes retain 72.1% capacity over 2500 cycles at 3 C, and those with NCM811 cathodes maintain 82.5% capacity after 500 cycles at 1 C. This work reveals that dipole-anchoring of residual solvent transforms a detrimental component into a stabilizer, offering an effective strategy to simultaneously enhance ion transport and interfacial stability in polymer-based solid-state lithium batteries.

CRediT authorship contribution statement

Chao Wang: Methodology. **Haitao Huang:** Writing – review & editing. **Qian Cheng:** Writing – original draft, Investigation, Conceptualization. **Shengjie Xia:** Investigation. **Yao Liu:** Methodology. **Zezhou Lin:** Visualization. **Ke Fan:** Methodology. **Hongwei Chen:** Methodology. **Changhong Wang:** Writing – review & editing. **Yipeng Sun:** Methodology. **Jing Zhang:** Investigation. **Xueliang Sun:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nanoen.2026.112343.

Data Availability

Data will be made available on request.

References

[1] X.Y. Huang, C.Z. Zhao, W.J. Kong, N. Yao, Z.Y. Shuang, P. Xu, S. Sun, Y. Lu, W. Z. Huang, J.L. Li, L. Shen, X. Chen, J.Q. Huang, L.A. Archer, Q. Zhang, Tailoring polymer electrolyte solvation for 600 Wh kg^{-1} lithium batteries, Nature 646 (2025) 343–350, https://doi.org/10.1038/s41586-025-09565-z.

[2] X. Yu, R. Chen, L. Gan, H. Li, L. Chen, Battery Safety: From Lithium-Ion to Solid-State Batteries, Engineering 21 (2023) 9–14, https://doi.org/10.1016/j.eng.2022.06.022.

[3] Z. Lin, W. Yuan, H. Huang, Pushing the boundaries of high-performance Li-ion batteries: manufacturing for harsh environments, Int. J. Extrem. Manuf. 8 (2026) 033002, https://doi.org/10.1088/2631-7990/ae42a0.

[4] P. Lennartz, B.A. Paren, A. Herzog-Arbeitman, X.C. Chen, J.A. Johnson, M. Winter, Y. Shao-Horn, G. Brunklaus, Practical considerations for enabling Li|polymer electrolyte batteries, Joule 7 (2023) 1471–1495, https://doi.org/10.1016/j.joule.2023.06.006.

[5] Z. Wang, L. Shen, S. Deng, P. Cui, X. Yao, 10 μm-thick high-strength solid polymer electrolytes with excellent interface compatibility for flexible all-solid-state lithium-metal batteries, Adv. Mater. 33 (2021) e2100353, https://doi.org/10.1002/adma.202100353.

[6] Y. Liu, P. Wang, Z. Yang, L. Wang, Z. Li, C. Liu, B. Liu, Z. Sun, H. Pei, Z. Lv, W. Hu, Y. Lu, G. Zhu, Lignin derived ultrathin all-solid polymer electrolytes with 3D single-ion nanofiber ionic bridge framework for high performance lithium batteries, Adv. Mater. 36 (2024) e2400970, https://doi.org/10.1002/adma.202400970.

[7] S. Xia, X. Zhang, Z. Jiang, X. Wu, J.A. Yuwono, C. Li, C. Wang, G. Liang, M. Li, F. Zhang, Y. Yu, Y. Jiang, J. Mao, S. Zheng, Z. Guo, Ultrathin polymer electrolyte with fast ion transport and stable interface for practical solid-state lithium metal batteries, Adv. Mater. 37 (2025) e2510376, https://doi.org/10.1002/adma.202510376.

[8] L. Ren, P. Zou, L. Wang, Y. Jing, H.L. Xin, Effect of vinylene carbonate additive in polyacrylate-based polymer electrolytes for high-voltage lithium-metal batteries, Energy Mater. Devices 2 (2024) 9370049, https://doi.org/10.26599/emd.2024.9370049.

[9] T.-H.L.Ying Wei, Wenjiang Zhou, Hang Cheng, Xueting Liu, Jia Kong, Yue Shen, Henghui Xu, Yunhui Huang, Enabling all-solid-state Li metal batteries operated at 30 ℃ by molecular regulation of polymer electrolyte, Adv. Energy Mater. 13 (2023) 2203547, https://doi.org/10.1002/aenm.202203547.

[10] X. Su, X.-P. Xu, Z.-Q. Ji, J. Wu, F. Ma, L.-Z. Fan, Polyethylene oxide-based composite solid electrolytes for lithium batteries: current progress, low-temperature and high-voltage limitations, and prospects, Electrochem. Energy Rev. 7 (2024) 1–38, https://doi.org/10.1007/s41918-023-00204-7.

[11] J. Mi, J. Yang, L. Chen, W. Cui, Y. Li, X. An, J. Ma, K. Yang, Y. Xie, J. Biao, Y. Long, H. Ge, B. Han, R. Ke, G. Xiao, S. Tan, D. Zhang, X. Cheng, T. Hou, Y.F. Huang, M. Liu, W. Lv, L. Gan, Y.B. He, Q.H. Yang, F. Kang, A ductile solid electrolyte interphase for solid-state batteries, Nature 647 (2025) 86–92, https://doi.org/10.1038/s41586-025-09675-8.

[12] L. Chen, T. Gu, J. Mi, Y. Li, K. Yang, J. Ma, X. An, Y. Jiang, D. Zhang, X. Cheng, S. Guo, Z. Han, T. Hou, Y. Cao, M. Liu, W. Lv, Y.B. He, F. Kang, Homogeneous polymer-ionic solvate electrolyte with weak dipole-dipole interaction enabling long cycling pouch lithium metal battery, Nat. Commun. 16 (2025) 3517, https://doi.org/10.1038/s41467-025-58689-3.

[13] Y.L. Liao, X.L. Wang, H. Yuan, Y.J. Li, C.M. Xu, S. Li, J.K. Hu, S.J. Yang, F. Deng, J. Liu, J.Q. Huang, Ultrafast Li-rich transport in composite solid-state electrolytes, Adv. Mater. 37 (2025) e2419782, https://doi.org/10.1002/adma.202419782.

[14] Q. Liu, G. Yang, X. Li, S. Zhang, R. Chen, X. Wang, Y. Gao, Z. Wang, L. Chen, Polymer electrolytes based on interactions between [solvent-Li^+] complex and solvent-modified polymer, Energy Storage Mater. 51 (2022) 443–452, https://doi.org/10.1016/j.ensm.2022.06.040.

[15] D. Zhang, Y. Liu, D. Li, S. Li, Q. Xiong, Z. Huang, S. Wang, H. Hong, J. Zhu, H. Lv, C. Zhi, Salt dissociation and localized high-concentration solvation at the interface of a fluorinated gel and polymer solid electrolyte, Energy Environ. Sci. 18 (2025) 227–235, https://doi.org/10.1039/d4ee04078c.

[16] L. Yang, Y. Chu, Y. Feng, Y. Mu, L. Zou, C. Li, C. Huang, H. Gu, C. Li, Q. Zhang, L. Zeng, Breaking voltage limitations: triethyl phosphate-engineered PVDF-based electrolytes with dual-interphase stabilization for 4.7 V-class Quasi-solid-state lithium metal batteries, J. Am. Chem. Soc. 147 (2025) 25940–25949, https://doi.org/10.1021/jacs.5c08493.

[17] X. Zhang, J. Han, X. Niu, C. Xin, C. Xue, S. Wang, Y. Shen, L. Zhang, L. Li, C. W. Nan, High cycling stability for solid-state Li metal batteries via regulating solvation effect in poly(vinylidene fluoride)-based electrolytes, Batter. Supercaps 3 (2020) 876–883, https://doi.org/10.1002/batt.202000081.

[18] X. Yu, L. Zhao, Y. Li, Y. Jin, D.J. Politis, H. Liu, H. Wang, M. Liu, Y.-B. He, L. Wang, Weakening ionic coordination for high ionic conductivity composite solid electrolytes, ACS Energy Lett. 9 (2024) 2109–2115, https://doi.org/10.1021/acsenergylett.4c00636.

[19] L. Niu, R. Wei, Y. Zhang, J. You, M. Wübbenhorst, Bromide-driven reorganization of lithium solvation shells enables dynamically decoupled ion transport and interfacial stability in semi-solid polymer electrolytes for lithium metal batteries, J. Energy Chem. 113 (2026) 531–540, https://doi.org/10.1016/j.jechem.2025.09.053.

[20] K. Yang, J. Ma, Y. Li, J. Jiao, S. Jiao, X. An, G. Zhong, L. Chen, Y. Jiang, Y. Liu, D. Zhang, J. Mi, J. Biao, B. Li, X. Cheng, S. Guo, Y. Ma, W. Hu, S. Wu, J. Zheng, M. Liu, Y.B. He, F. Kang, Weak-interaction environment in a composite electrolyte enabling ultralong-cycling high-voltage solid-state lithium batteries, J. Am. Chem. Soc. (2024), https://doi.org/10.1021/jacs.4c00976.

[21] W. Yang, Y. Liu, X. Sun, Z. He, P. He, H. Zhou, Solvation-tailored PVDF-based solid-state electrolyte for high-voltage lithium metal batteries, Angew. Chem. Int. Ed. 63 (2024) e202401428, https://doi.org/10.1002/anie.202401428.

[22] X. Chang, R. Cheng, T. Wang, X. Kan, M. Jia, Z. Bi, X. Fan, X. Guo, Fluorinated carbon nitride-assisted solvation-regulation engineering toward polyvinylidene fluoride-based electrolytes for long-lifespan solid-state lithium batteries, Energy Environ. Sci. 18 (2025) 9490–9501, https://doi.org/10.1039/d5ee03323c.

[23] J. Lv, Y. Li, K. Yang, X. Liu, Y. Dou, Z. Zhang, D. Zhang, P. Shi, M. Liu, Y.-B. He, ZIF-based heterojunction filler enhancing Li-ion transport of composite solid-state electrolytes, Energy Mater. Devices 3 (2025) 9370063, https://doi.org/10.26599/emd.2025.9370063.

[24] M. Li, H. An, Y. Song, Q. Liu, J. Wang, H. Huo, S. Lou, J. Wang, Ion-dipole-interaction-induced encapsulation of free residual solvent for long-cycle solid-state lithium metal batteries, J. Am. Chem. Soc. 145 (2023) 25632–25642, https://doi.org/10.1021/jacs.3c07482.

[25] D. Zhang, Y. Liu, S. Yang, J. Zhu, H. Hong, S. Li, Q. Xiong, Z. Huang, S. Wang, J. Liu, C. Zhi, Inhibiting residual solvent induced side reactions in vinylidene fluoride-based polymer electrolytes enables ultra-stable solid-state lithium metal batteries, Adv. Mater. 36 (2024) e2401549, https://doi.org/10.1002/adma.202401549.

[26] Y. Li, W. Yuan, Z. Hu, Y. Shen, G. Wu, F. Cong, X. Fu, F. Lu, Y. Li, P. Liu, Y. Huang, J. Li, Constructing PVDF-based polymer electrolyte for lithium metal batteries by polymer-induced phase structure adjustment strategy, Adv. Funct. Mater. 35 (2025) 2424763, https://doi.org/10.1002/adfm.202424763.

[27] D. Dai, P. Yan, D. Liu, Z. Zhang, Y. Chen, H. Li, H. Zhu, Z. Gu, J. Guo, B. Li, X. Wu, Hydrogen bonds boost lithium salt dissociation in composite solid-state electrolyte: enhanced cycling life of lithium metal batteries, Adv. Funct. Mater. 35 (2025) 2503696, https://doi.org/10.1002/adfm.202503696.

[28] C. Xue, X. Zhang, S. Wang, L. Li, C.-W. Nan, Organic-organic composite electrolyte enables ultralong cycle life in solid-state lithium metal batteries, ACS Appl. Mater. Interfaces 12 (2020) 24837–24844, https://doi.org/10.1021/acsami.0c05643.

[29] M. Li, T. Tian, X. Yang, Y. He, D. Zhang, P. Müller-Buschbaum, S. Yang, B. Li, Analogue molecular doping engineering enables high ionic conductivity of polyvinylidene fluoride-based polymer electrolytes, ACS Nano 19 (2025) 20084–20095, https://doi.org/10.1021/acsnano.5c04141.

[30] Q. Wang, S. Wang, T. Lu, L. Guan, L. Hou, H. Du, H. Wei, X. Liu, Y. Wei, H. Zhou, Ultrathin solid polymer electrolyte design for high-performance Li metal batteries: a perspective of synthetic chemistry, Adv. Sci. 10 (2022) e2205233, https://doi.org/10.1002/advs.202205233.

[31] J. Zhang, Y. Zeng, Q. Li, Z. Tang, D. Sun, D. Huang, L. Zhao, Y. Tang, H. Wang, Polymer-in-salt electrolyte enables ultrahigh ionic conductivity for advanced solid-state lithium metal batteries, Energy Storage Mater. 54 (2023) 440–449, https://doi.org/10.1016/j.ensm.2022.10.055.

[32] D. Zhang, Y. Liu, D. Li, S. Li, Q. Xiong, Z. Huang, S. Wang, H. Hong, J. Zhu, C. Zhi, Modulating salt dissociation and solvent immobilization through dipole interactions in polymer electrolytes for lithium metal batteries, Adv. Mater. 38 (2025) e12960, https://doi.org/10.1002/adma.202512960.

[33] J. Hou, W. Xie, L. Shang, S. Wu, Y. Cui, Y. Li, Z. Yan, K. Zhang, Y. Lu, J. Chen, Nanocrystals with aggregate anionic structure enable ion transport decoupling of chain segment movement in poly(ethylene oxide) electrolytes, Angew. Chem. Int. Ed. 64 (2025) e202418783, https://doi.org/10.1002/anie.202418783.

[34] W. Huang, G. Zhao, B. Zhang, T. Li, H. Zhang, J. Wang, X. Zhao, X. Hu, Y. Xu, Recent advances for cation-anion aggregates in solid polymer electrolytes: mechanism, strategies, and applications, Small Methods 9 (2025) e2401998, https://doi.org/10.1002/smtd.202401998.

[35] L. Mathies, D. Diddens, D. Dong, D. Bedrov, H. Leipner, Transport mechanism of lithium ions in non-coordinating P(VdF-HFP) copolymer matrix, Solid. State Ion. 357 (2020) 115497, https://doi.org/10.1016/j.ssi.2020.115497.

[36] Y. Zhao, L. Li, D. Zhou, Y. Ma, Y. Zhang, H. Yang, S. Fan, H. Tong, S. Li, W. Qu, Opening and constructing stable lithium-ion channels within polymer electrolytes, Angew. Chem. Int. Ed. 63 (2024) e202404728, https://doi.org/10.1002/anie.202404728.
[37] Y. Lu, C.-Z. Zhao, J.-Q. Huang, Q. Zhang, The timescale identification decoupling complicated kinetic processes in lithium batteries, Joule 6 (2022) 1172–1198, https://doi.org/10.1016/j.joule.2022.05.005.
[38] Q. Cheng, K. Fan, J.M. Cao, Z. Lin, J. Fu, S. Xia, C. Wang, Y. Liu, S. Zhang, C. Wang, X. Sun, H. Huang, Biomimetic ion channel design for simultaneous lithium-ion flux regulation and interfacial stabilization in lithium metal batteries, Small (2026) e74056, https://doi.org/10.1002/smll.74056.
[39] K. Yang, L. Chen, J. Ma, C. Lai, Y. Huang, J. Mi, J. Biao, D. Zhang, P. Shi, H. Xia, G. Zhong, F. Kang, Y.B. He, Stable interface chemistry and multiple ion transport of composite electrolyte contribute to ultra-long cycling solid-state $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$/lithium metal batteries, Angew. Chem. Int. Ed. 60 (2021) 24668–24675, https://doi.org/10.1002/anie.202110917.
[40] B. Han, Z. Zhang, Y. Zou, K. Xu, G. Xu, H. Wang, H. Meng, Y. Deng, J. Li, M. Gu, Poor stability of Li_2CO_3 in the solid electrolyte interphase of a lithium-metal anode revealed by cryo-electron microscopy, Adv. Mater. 33 (2021) e2100404, https://doi.org/10.1002/adma.202100404.
[41] W. Li, M. Li, S. Wang, P.H. Chien, J. Luo, J. Fu, X. Lin, G. King, R. Feng, J. Wang, J. Zhou, R. Li, J. Liu, Y. Mo, T.K. Sham, X. Sun, Superionic conducting vacancy-rich beta-Li_3N electrolyte for stable cycling of all-solid-state lithium metal batteries, Nat. Nanotechnol. 20 (2025) 265–275, https://doi.org/10.1038/s41565-024-01813-z.
[42] S. Wu, W. Feng, P. Wang, C. Ma, L. Shen, Q. Shi, Y. Liu, Y. Zhang, X. Zhu, J. Zhang, Z. Jiao, Y. Zhao, Kinetics-reinforced mechano-electrochemically stable lithium-silicon-nitrogen composites toward dendrite-free all-solid-state batteries, Adv. Energy Mater. 15 (2025) e04107, https://doi.org/10.1002/aenm.202504107.
[43] Y. Wang, P. Yuan, X.X. Liu, S. Feng, M. Cao, J. Ding, J. Liu, S.Z. Kure-Chu, T. Hihara, L. Pan, Z. Sun, Sacrificial NH_4HCO_3 inhibits fluoropolymer/garnet interfacial reactions Toward 1 mS cm^{-1} and 5V-level composite solid electrolyte, Adv. Funct. Mater. 34 (2024) 2405060, https://doi.org/10.1002/adfm.202405060.
[44] L. Liu, Y. Shi, M. Liu, Q. Zhong, Y. Chen, B. Li, Z. Li, T. Zhang, H. Su, J. Peng, N. Yang, P. Wang, A. Fisher, J. Niu, F. Wang, An Ultrathin Solid Electrolyte for High-Energy Lithium Metal Batteries, Adv. Funct. Mater. 34 (2024) 2403154, https://doi.org/10.1002/adfm.202403154.
[45] J. Shan, R. Gu, J. Xu, S. Gong, S. Guo, Q. Xu, P. Shi, Y. Min, Heterojunction ferroelectric materials enhance ion transport and fast charging of polymer solid electrolytes for lithium metal batteries, Adv. Energy Mater. 15 (2025) 2405220, https://doi.org/10.1002/aenm.202405220.
[46] Y. Shan, L. Li, X. Chen, S. Fan, H. Yang, Y. Jiang, Gentle haulers of lithium-ion-nanomolybdenum carbide fillers in solid polymer electrolyte, ACS Energy Lett. 7 (2022) 2289–2296, https://doi.org/10.1021/acsenergylett.2c00849.
[47] W. Yang, Z. Zhang, X. Sun, Y. Liu, C. Sheng, A. Chen, P. He, H. Zhou, Tailoring the electrode-electrolyte interface for reliable operation of all-climate 4.8 V Li|| NCM811 batteries, Angew. Chem. Int. Ed. 63 (2024) e202410893, https://doi.org/10.1002/anie.202410893.

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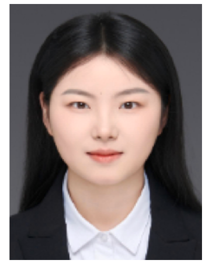

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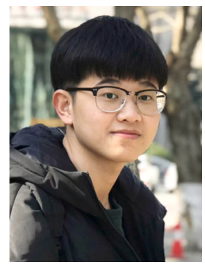

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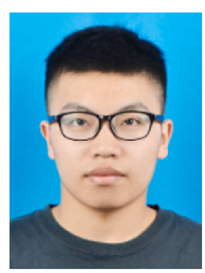

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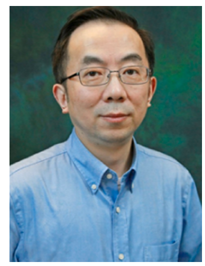

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